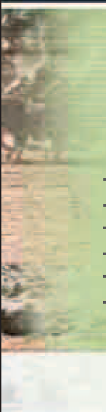
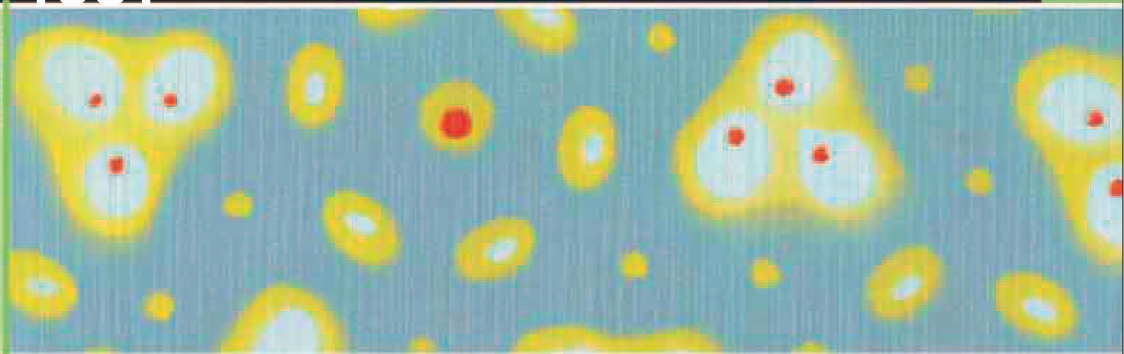
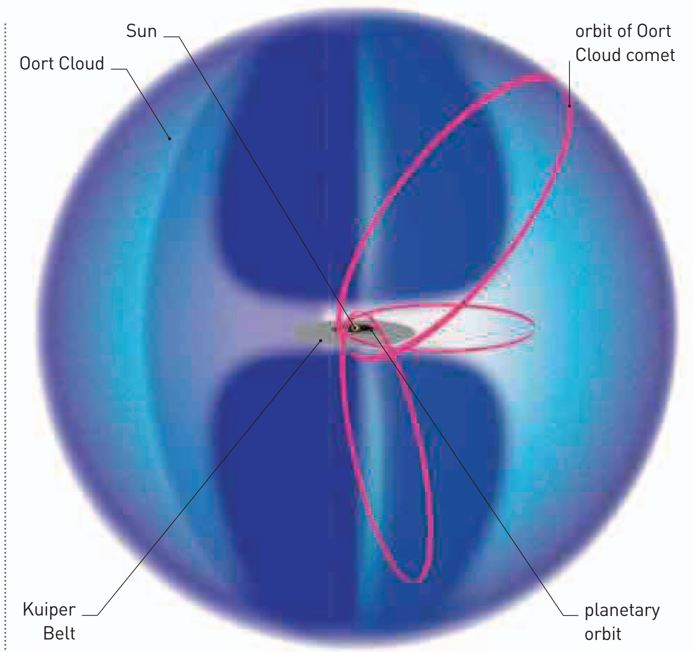


600,000,000

PEAK NUMBER OF **RABBITS** IN **AUSTRALIA BEFORE** INTRODUCTION OF **MYXOMA**



Science influenced fashion at the 1951 Festival of Britain, where fabric and wallpaper designs based on X-ray crystallography (see 1945) were exhibited.



Oort Cloud
Made up of billions of comets, the Oort Cloud marks the hypothetical outer boundary of the Solar System. Orbits of Oort Cloud comets are more than a thousand times bigger than planetary orbits.

eradicated, their numbers never recovered to pre-1950s levels. In the 1970s, Fenner went on to use his skills in disease control and played a crucial role in the World Health Organization's successful **global elimination of human smallpox**.

The start of a new decade also saw an effort to bring into focus the most mysterious part of Earth's surface: the **ocean floor**. Geologists such as **Marie Tharp** (1920–2006) and **Bruce Heezen** (1924–77) had used photographic methods to locate sunken aircraft from World War II. They went on to **map the underwater seascape**—discovering submerged mountain ranges along the way.

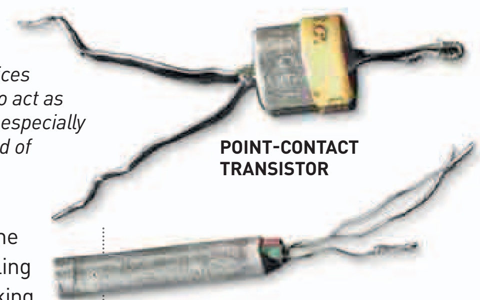
In this year, Dutch astronomer and physicist **Jan Oort** (1900–92) suggested that **comets came from a cloudlike reservoir** at the edge of the Solar System. Modern astronomers believe that Oort was right.

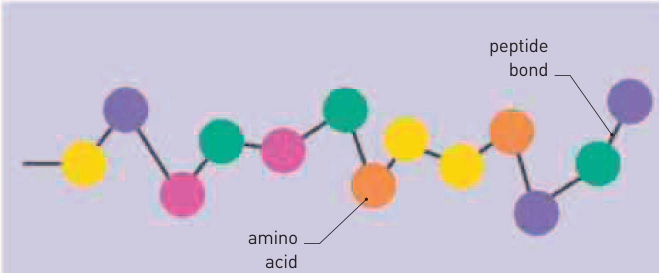
IN 1951, the *Lancet*, a medical journal, published an article by British physician **Richard Asher** (1912–69) about a new mental disorder in which sufferers sought attention by fabricating medical ailments. Asher called it the **Münchhausen syndrome**—after the 18th-century German baron who invented wild stories about his life that he claimed were true.

The same year also saw an important breakthrough in the field of molecular biology. Unraveling the molecular structure of complex biological substances such as proteins had been one of the challenges of analytical chemistry. British biochemist **Fred Sanger** (b.1918) studied one particular protein, **insulin**, which consisted of interlocked chains of smaller variable components called amino acids. By chemically splitting these chains, Sanger was able to determine the **types of amino acids**, and even the sequence in which they were linked together. Sanger was the first scientist to show that this **sequence was the same for all insulin** molecules, and that **different kinds of proteins** have **unique amino acid sequences**. It would take scientists more than a decade to fully appreciate the implications of Sanger's findings: that in the living body, individual genes in the DNA hold the instructions for assembling particular proteins by linking amino acids together in the correct order.

On July 4, American inventor **William Shockley** (1910–89), working at Bell Telephone Laboratories in New Jersey, announced the invention of the **junction transistor**. Shockley and his team had made their first transistor in 1947, but it was the improved design of 1951 that would become the standard component of electronic devices for the next 30 years.

Early transistors
Transistors revolutionized the world of electronic devices and circuitry. Their ability to act as switches also proved to be especially valuable in the growing field of computer technology.





AMINO ACID CHAINS

Protein molecules perform critical roles in the bodies of living things, such as driving metabolism and helping cells absorb nutrients. In 1951, Fred Sanger found that the chainlike molecule of a certain kind of protein—insulin—consisted of a unique sequence of amino acids. He also found that different kinds of proteins have different sequences—this determines how each chain folds into a shape for a particular purpose.

September Australian microbiologist **Frank Fenner** introduces **myxomatosis** in Australia to control the rabbit population

October British mathematician **Alan Turing** describes **artificial intelligence**

German biologist **Willi Hennig** publishes the first description of the method of **biological cladistics**

Canadian biomedical engineer **John Hopps** designs first **external heart pacemaker**

February **Richard Asher** describes **Münchhausen syndrome** in the *Lancet*

February First **commercial electronic computer** (Mark 1), developed by the Ferranti company

March **Edward Teller** and **Stanislaw Ulam** create the concept design for first **workable thermonuclear bomb**

July 4 American electronic engineer **William Shockley** announces invention of first **junction transistor**

Dutch zoologist and animal psychologist **Niko Tinbergen** publishes *The Study of Instinct*

September British biochemist **Fred Sanger** publishes first **amino acid sequence** of a protein (insulin)

September 25 American inventor William Shockley **obtains patent** for his junction transistor